Climate Shocks and the Credit Market: A Monetary Policy Transmission Lens

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1 Introduction

According to Bernanke and Gertler (1995), the credit channel is one of the key monetary policy transmission channels. Monetary policy affects the credit channel through its impact on the external finance premium [1]. Two linkages underscore monetary policy impacts on this premium: the balance sheet and the bank lending channel. The balance sheet channel stems from dynamics in the balance sheet and the incomes of borrowers, while the lending channel deals with how interest rates could impact lenders' loan supply volumes (Hall, 2001). The balance sheet channel has been well established as a significant monetary policy transmission channel (Aysun and Hepp, 2013; Altavilla et al., 2024; Ashcraft and Campello, 2007). Notwithstanding, there remains room to provide empirical estimations of the credit channel's strength due to recent developments in the identification of monetary policy shocks. This is especially true for emerging economies where this work is scant (Pirozhkova and Viegi, 2025). This paper hones in on the balance sheet channel by estimating how climate shocks can reduce (or indeed enhance) the effectiveness of unanticipated monetary policy decisions on the behaviour of borrowers.

The main motivation of this paper is that like monetary policy shocks, climate shocks – because of their impacts on capital and labour inputs and, in turn, their erosion of firm profitability – can cause frictions within the credit market and have consequential financial stability ramifications (Dafermos et al., 2018; Aglietta and Espagne, 2016). In so doing, climate shocks could widen the external finance premium and, in turn, stymie the efficacy of monetary policy pass-through. While this seems a likely course, there is also the potential for monetary policy to mute the deleterious climate shock effects on the credit market, in line with the literature that that studies how monetary policy can be used as a tool to tackle climate change (Roy, 2024; Monasterolo and Raberto, 2018). In either case, climate shocks play a role in the functioning of the credit market and, consequently, the credit market monetary transmission channel.

I will investigate to what extent the manifestation of climate shocks act as a credit market

[1] The external finance premium is governed by credit-market frictions which cause departures from lenders expected return and potential borrowers' expected costs.

friction and how this (potential) friction impacts monetary policy transmission. This involves three parts. First, I will compute the monetary policy shock variable. Second, I will estimate the causal relationships of interest in this work using a panel regression. Finally, I will use a quantile regression procedure to nuance these results.

Only physical shocks, not transition

2 Literature review

3 Data and methodology

3.1 Data

The analysis draws on three main data sources: climate shocks, monetary policy shocks, and credit extension data. The sample covers South Africa over the period 2011–2025. All series are observed at a quarterly frequency, determined by the availability of the monetary policy shock data; higher-frequency series are aggregated to quarterly values by taking averages or sums where appropriate.

Climate data: Temperature and precipitation data are sourced from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts[2]. ERA5 provides globally gridded, hourly atmospheric estimates at a 31 km horizontal resolution. Grid-level values are weighted by population and aggregated to monthly averages over South Africa's land area, from which we construct climate shock variables that capture deviations from long-run historical norms in temperature and precipitation.

The climate shocks are calculated as follows:

$$W_t = \frac{T_t - \overline{T}_t}{\sigma(T_t)}, \tag{1}$$

where W_t is the temperature or precipitation shock at time t and T_t is the observed population-weighted climate variable. For both temperature and precipitation, $\overline{T}_t$ is the calendar-month

[2]see https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview

mean computed over the full sample period (e.g., the mean of all January values from 2011 to 2025), and $\sigma(T_t)$ is the corresponding within-month standard deviation. The resulting shock series are shown in Figure A3 in the Appendix.

Monetary policy data: The monetary policy shock series is constructed using high-frequency policy surprises for South Africa from Pirozhkova and Viegi (2025), alongside the repurchase (repo) rate from the South African Reserve Bank (SARB). The high-frequency surprises serve as an instrument to purge the endogenous component of policy rate changes, yielding a shock series that satisfies the exogeneity requirement of the regression framework described in Section 3.2. For robustness, we consider both market-based shocks from Pirozhkova and Viegi (2025) (see Figure A4) and a theory-based shock derived from the policy rate, as described in Section 3.2 below (see Figure A5). Surprises are timed to coincide with policy rate announcement dates and aggregated to a quarterly frequency. Underlying the calculation of the are the forecasts of the real GDP growth, the policy rate (repo-rate), and the inflation rate from Bloomberg Consensus.

Credit extension data: Private sector credit extension data — disaggregated by borrower type (households and corporates) at the bank level — are obtained from the SARB's BA900 and BA930 bank compliance reports.[3] These series include both credit volumes and lending rates, and are constructed following Sibande et al. (2025) and Sibande and Milne (2025) by sorting and aggregating individual bank records into the broad credit categories used in the study: unsecured, secured, and mortgage credit. Summary charts are provided in Figure A1 and Figure A2.

3.2 Methodology

Structural shocks: Rather than using the policy interest rate directly — which responds to prevailing economic conditions and is therefore endogenous — we use an unexpected change in monetary policy, commonly called a monetary policy shock, to isolate the causal effect of policy on credit markets (Miyajima, 2024). We construct this shock following Merrino (2022),

[3]BA930 data at the bank level are not publicly available.

who adapts the approach of Romer and Romer (2004) to the South African context. The idea is to regress the change in the repo rate forecast on variables that policymakers were already tracking, and treat whatever is left over — the part of the rate change that cannot be explained by those forecasts — as the true policy surprise. Formally:

$$\Delta r_t = \alpha + \tilde{r}_t + \sum_{t=0}^{2} \beta_t \Delta \tilde{y}_t + \sum_{t=0}^{2} \delta_t \tilde{\pi}_t + \varepsilon_t, \tag{2}$$

where Δr_t is the change in the repo rate forecast, $\tilde{r}_t$ is the forecasted repo rate, $\tilde{y}_t$ is the forecasted real GDP growth rate, and $\tilde{\pi}_t$ is the forecasted inflation rate. The residual ε_t — the unexplained part of the forecast revision — is our monetary policy shock, MPShock_t.[4]

As an alternative, we construct a second measure of the monetary policy surprise following Miyajima (2024). This approach uses forecast errors — the gap between what forecasters expected and what actually occurred — rather than SARB forecasts. The construction proceeds in two steps. First, for each variable i, the forecast error is:

$$FE_t^i = ACT_t^i - EXP_t^i, \tag{3}$$

where $i \in \{\text{policy rate}, \text{inflation}, \text{GDP growth}\}$, ACT_t^i is what was actually observed, and EXP_t^i is the Bloomberg consensus forecast made beforehand. In the second step, the policy rate forecast error is regressed on the inflation and output growth forecast errors:

$$FE_t^r = \alpha + \beta FE_t^\pi + \gamma FE_t^y + \epsilon_t, \tag{4}$$

where FE_t^π is the inflation forecast error and FE_t^y is the GDP growth forecast error. The residual ϵ_t is the monetary policy surprise — the part of the policy rate miss that has nothing to do with inflation or growth news, and is therefore not anticipated by market participants.

Market-based shocks: As a further robustness check, we use market-based monetary policy surprises built from a high-frequency identification approach (Gürkaynak et al., 2005). The core idea is to measure how financial market prices move in a narrow window around

[4] We include forecasts up to two quarters ahead to account for the common assumption among forecasters that policy is unchanged in the very near term. See Romer and Romer (2004) for details.

each MPC announcement: since little else happens in that window, any price change can be attributed to the policy news itself (Nakamura and Steinsson, 2018). Following Pirozhkova and Viegi (2025), we decompose the overall surprise into four components: a target factor (the surprise in the level of the policy rate), a forward guidance factor (the surprise in the expected future path of policy), a central bank information factor (what the central bank's communication revealed about the economic outlook that markets did not already know), and a country risk factor (shifts in sovereign risk perceptions around MPC dates). Each component is used in turn as an alternative measure of MPShock_t in Equation 6 and Equation 7.

Shock purging: Policymakers are aware of climate conditions and may already adjust rates in response to them. If so, our monetary policy shocks would partly reflect climate events rather than pure policy decisions, which would confound our estimates. To address this, we regress each monetary policy shock on the climate shock measures:

$$\text{MPShock}_t^i = \text{ClimateShock}_t^i + \varepsilon_t^i, \tag{5}$$

where MPShock_t^i is each of the monetary policy shocks described above and ClimateShock_t^i are the temperature and precipitation shocks. The residual ε_t^i — the part of the policy shock that cannot be explained by climate conditions — is retained as a "purged" shock. We use these purged shocks in all subsequent estimations.

Panel estimation: Our first estimating equation tests whether the effect of a monetary policy shock on credit is amplified or dampened when a climate shock occurs at the same time:

$$\begin{aligned} \text{Credit}_{s,t} =& \beta_0 + \beta_1 \text{Climate}_t^k + \beta_2 \text{MPShock}_t + \beta_3 (\text{Climate}_t^k \times \text{MPShock}_t) + \\ & \Gamma X_t + \gamma_s + \delta_t + \varepsilon_{s,t}, \end{aligned} \tag{6}$$

where $\text{Credit}_{s,t}$ is credit growth for agent s (households or corporates) at time t, and Climate_t^k captures climate shocks — temperature, precipitation, and extreme wet and dry events, indexed by k. The interaction term $\text{Climate}_t^k \times \text{MPShock}_t$ is the key coefficient of interest: a significant estimate would indicate that climate shocks alter how monetary policy transmits through credit markets. γ_s are agent fixed effects (absorbing time-invariant differences across borrower types), δ_t are time fixed effects, and $\varepsilon_{s,t}$ is the error term. X_t is a vector of controls added as a robustness check, including **GDP growth, the USD/ZAR exchange rate, the loan-to-deposit ratio, and the unemployment rate**.

Panel local projections: The panel regression above captures the contemporaneous relationship but cannot tell us how long any effect lasts. To trace the full dynamic response of credit over time, we use the local projections method of Jordà (2005), estimating a separate regression for each future horizon $h = 0, 1, \ldots, H$:

$$\begin{aligned} Credit_{s,t+h} - Credit_{s,t-1} =& \beta_0 + \beta_1 \text{Climate}_t^k + \beta_2 \text{MPShock}_t + \beta_3 (\text{Climate}_t^k \times \text{MPShock}_t) + \\ & \Gamma X_t + \gamma_s + \delta_t + \varepsilon_{s,t}, \end{aligned} \tag{7}$$

where $\text{Credit}_{s,t+h} - \text{Credit}_{s,t-1}$ is the cumulative change in credit growth from period $t-1$ to $t+h$, and h is the horizon in quarters. Plotting the estimated $\hat{\beta}_1$, and $\hat{\beta}_2 + \hat{\beta}_3$ across horizons gives the impulse response functions, showing the differential impact of climate on credit in the quarters following a shock. **All regressions use Driscoll-Kraay standard errors to account for correlation across agents and over time (Driscoll and Kraay, 1998)**.

4 Results

5 Drivers

5.1 Timing sensivity

5.2 Sectoral sensitivity

5.3 Type of shock

6 Conclusion

Table 1: Estimated coefficients from the fixed effects models

term	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
pop_temp_shock	1.052***											
miyajima_surprise	-0.764***											
pop_temp_shock:miyajima_surprise	1.361***											
pop_temp_shock		1.025***										
romer_surprise		0.107*										
pop_temp_shock:romer_surprise		0.533***										
pop_temp_shock			1.09***									
target			-0.013***									
pop_temp_shock:target			-0.016***									
pop_temp_shock				1.172***								
forward_guidance				0.024***								
pop_temp_shock:forward_guidance				-0.032***								
pop_temp_shock					1.01***							
central_bank_information					0.063***							
pop_temp_shock:central_bank_information					-0.041***							
pop_temp_shock						1.58***						
country_risk						-0.023***						
pop_temp_shock:country_risk						0.18***						
pop_precip_shock							-0.078***					
miyajima_surprise							1.142***					
pop_precip_shock:miyajima_surprise							0.413***					
pop_precip_shock								-0.054***				
romer_surprise								0.551***				
pop_precip_shock:romer_surprise								-0.104***				
pop_precip_shock									-0.028***			
target									-0.037***			
pop_precip_shock:target									0			
pop_precip_shock										-0.014**		
forward_guidance										0.034***		
pop_precip_shock:forward_guidance										0.005***		
pop_precip_shock											-0.032***	
central_bank_information											0.044***	
pop_precip_shock:central_bank_information											0.004***	
pop_precip_shock												-0.063***
country_risk												0.018***
pop_precip_shock:country_risk												0.037***

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A Appendix

A.1 Data

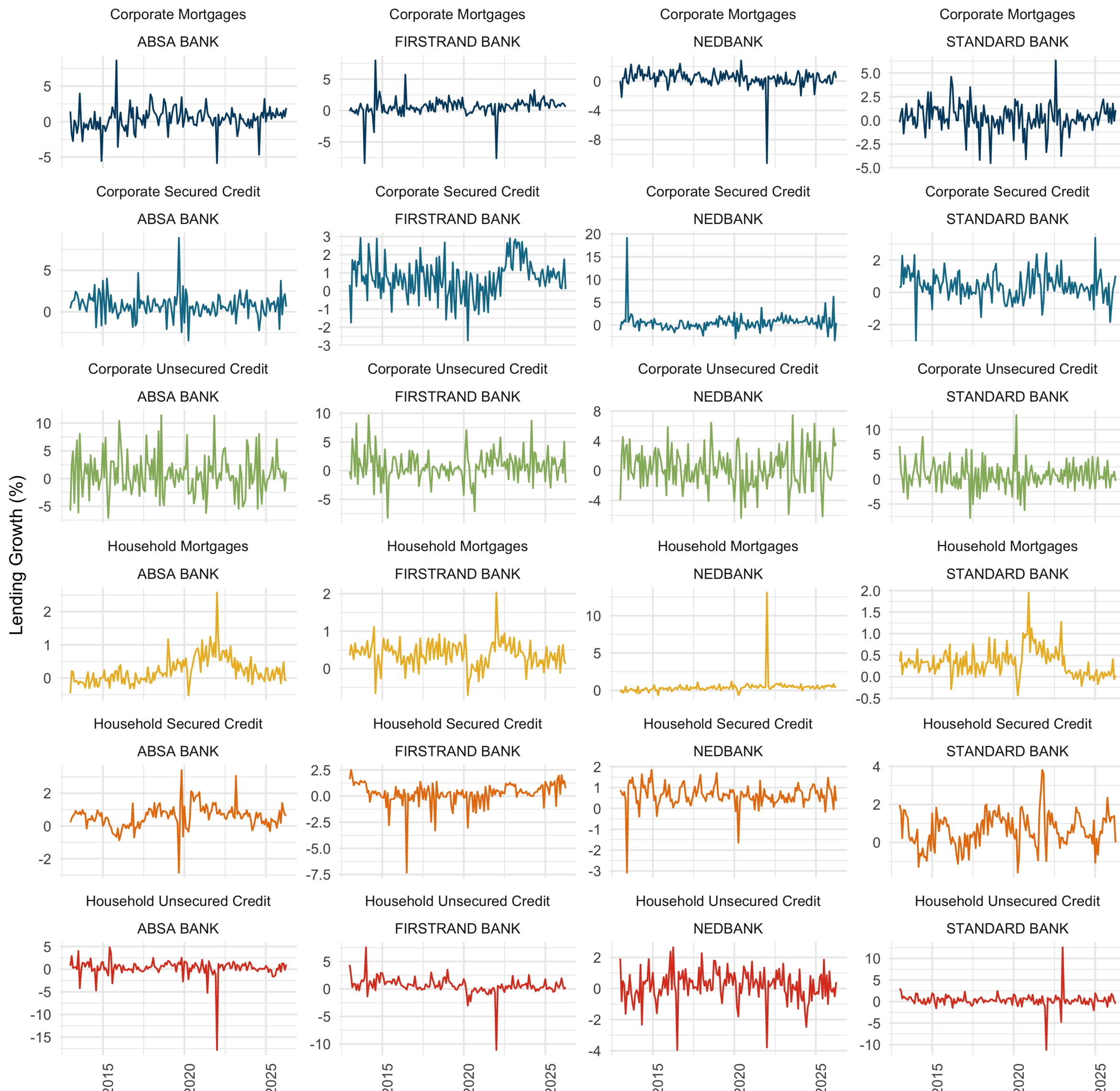

Figure A1: Lending volume growth over time

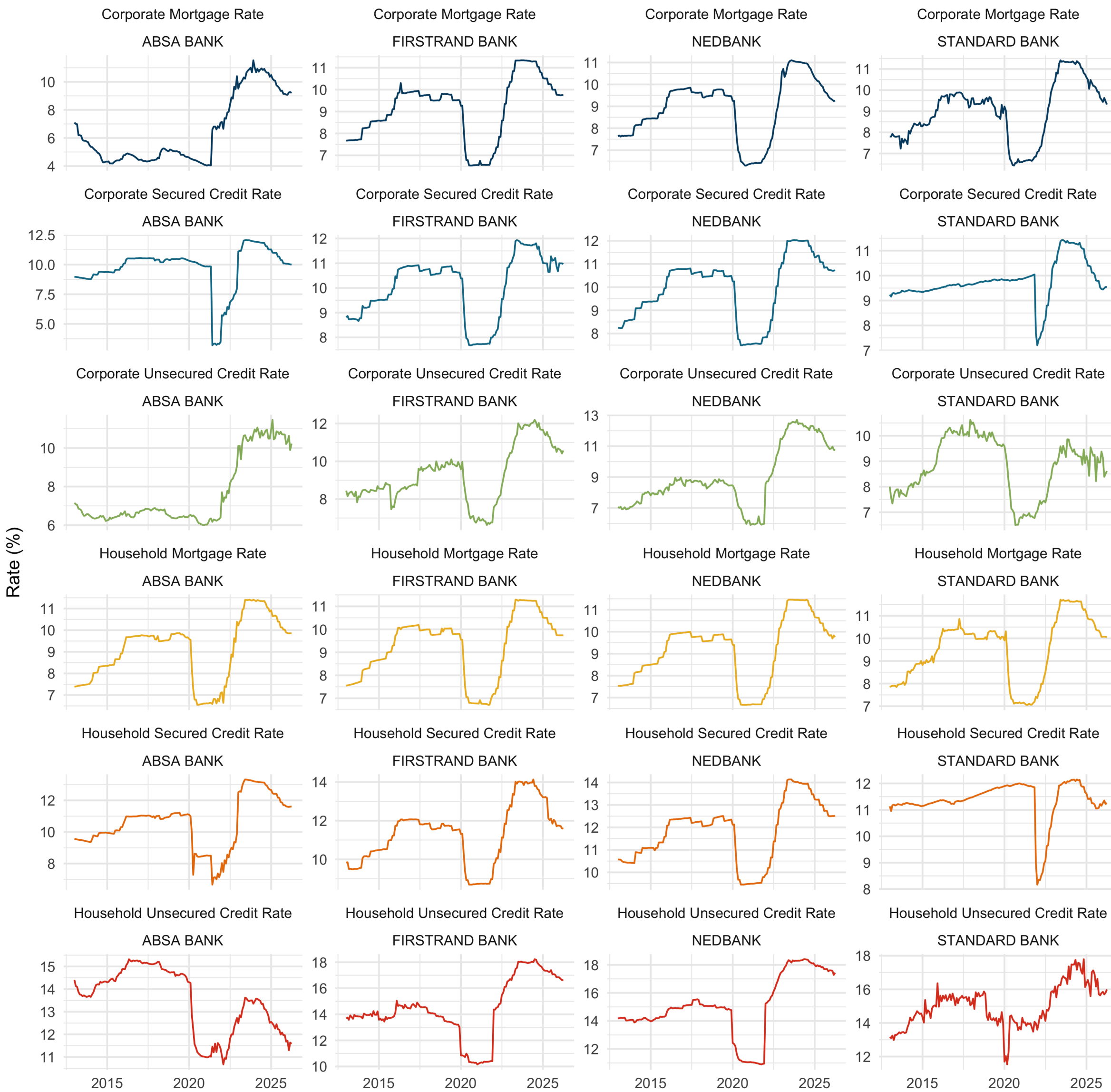

Figure A2: Lending rates overtime

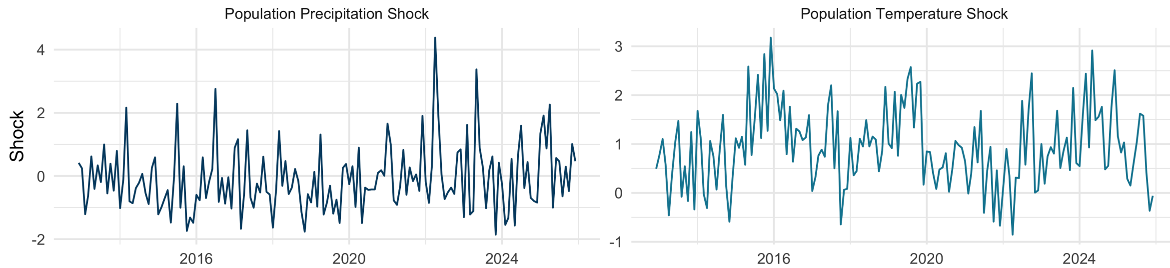

Figure A3: Climate Shocks Data

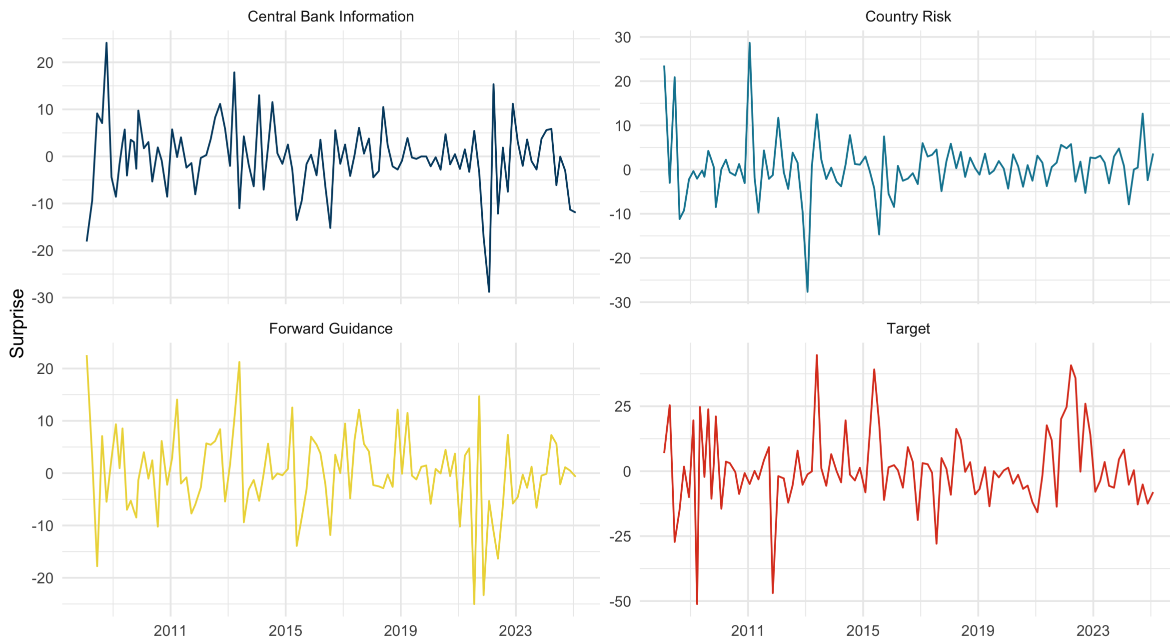

Figure A4: Market-Based Surprises Data

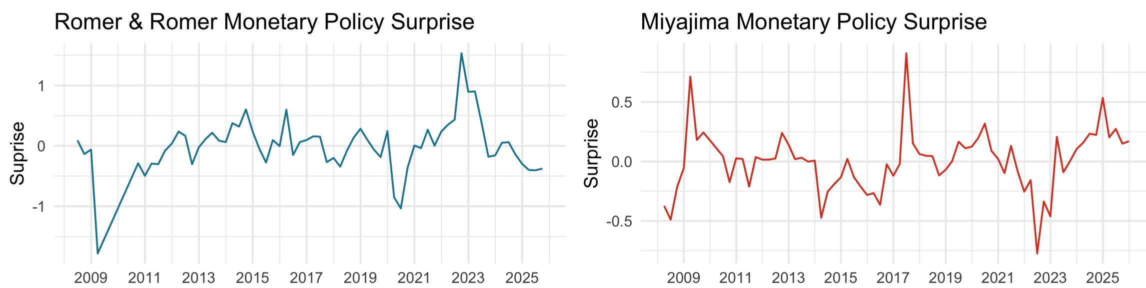

Figure A5: Theory-based surprises data

A.2 Descriptive Statistics

Table A1: Descriptive statistics

	Variable	Mean	Median	SD	Min	Max	N
Monetary Policy							
	Central Bank Information	-1.64	-0.42	5.69	-17.14	11.19	192
	Country Risk	0.64	0.47	3.38	-5.47	12.50	192
	Forward Guidance	-0.25	-0.47	6.93	-23.32	21.24	192
	Miyajima Surprise	-0.00	0.01	0.26	-0.77	0.91	208
	Romer Surprise	0.06	0.06	0.41	-1.04	1.53	204
	Target	3.12	0.64	13.46	-18.78	44.70	192
Corporate Credit							
	Corporate Mortgage Rate	8.43	9.07	2.04	4.06	11.35	212
	Corporate Mortgages	74.39	75.79	6.99	24.93	77.82	212
	Corporate Secured Credit	71.57	72.62	6.62	24.16	74.80	212
	Corporate Secured Credit Rate	9.91	9.99	1.36	3.30	12.06	212
	Corporate Unsecured Credit	76.25	77.16	7.01	25.81	79.22	212
	Corporate Unsecured Credit Rate	8.71	8.62	1.74	5.95	12.59	212
Household Credit							
	Household Mortgage Rate	9.30	9.75	1.41	6.56	11.67	212
	Household Mortgages	77.38	78.22	7.16	26.14	80.28	212
	Household Secured Credit	73.94	75.28	6.87	24.99	77.44	212
	Household Secured Credit Rate	11.28	11.35	1.42	7.09	14.11	212
	Household Unsecured Credit	73.93	75.12	6.86	24.58	76.35	212
	Household Unsecured Credit Rate	14.52	14.54	1.96	10.22	18.37	212
	Population Precipitation Shock	-0.26	-0.48	1.88	-4.54	6.38	204
	Population Precipitation Shock Abs Neg	1.36	1.12	0.90	0.00	4.54	204
	Population Precipitation Shock Pos	1.10	0.76	1.29	0.00	6.38	204
	Population Temperature Shock	0.95	1.06	0.60	-0.26	2.43	204
	Population Temperature Shock Abs Neg	0.04	0.00	0.09	0.00	0.42	204
	Population Temperature Shock Pos	0.99	1.06	0.55	0.05	2.43	204

A.3 Sector Credit Exposure

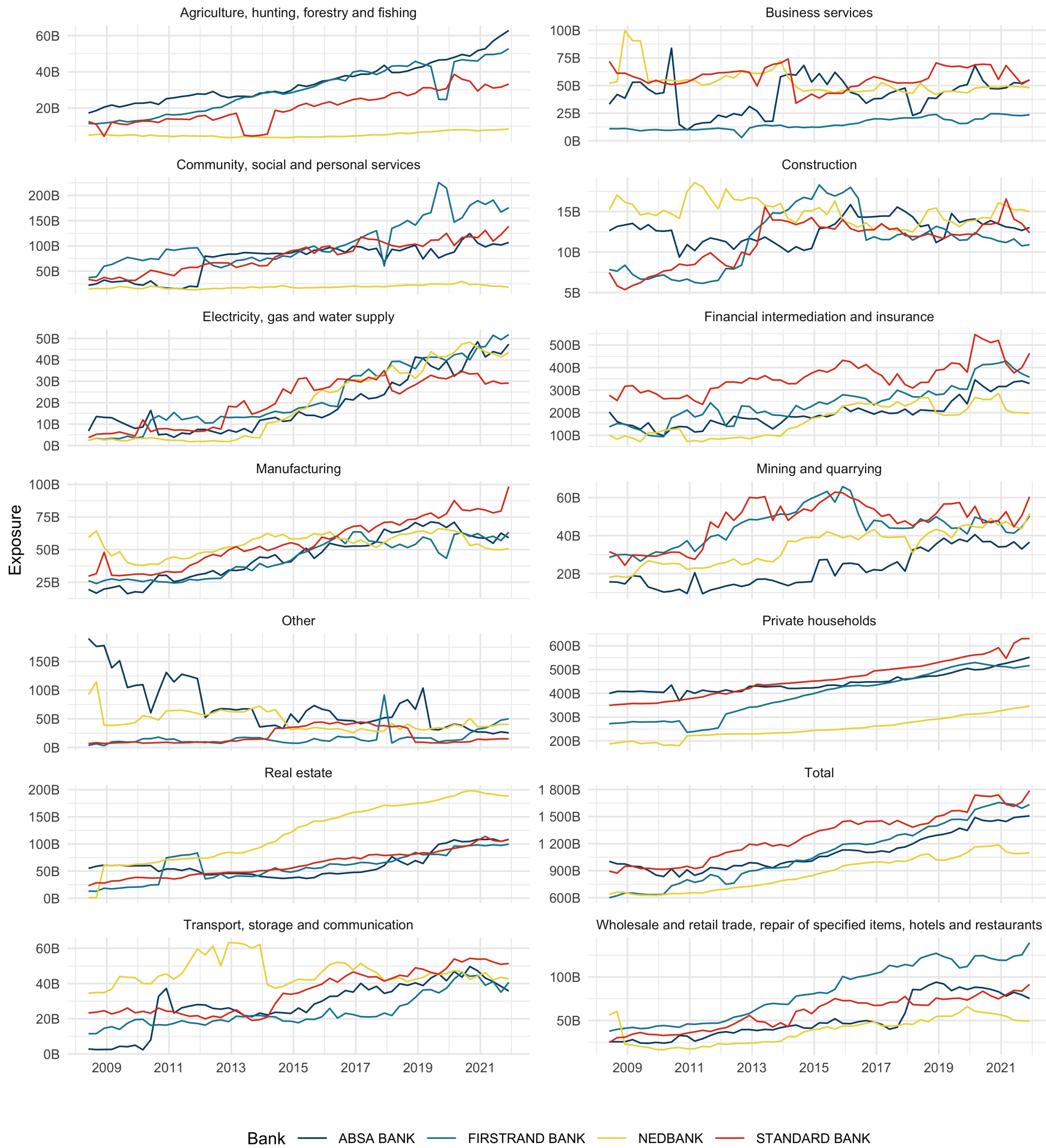

Figure A6: Sector credit exposure